



From Results to Discussion and back to Objectives

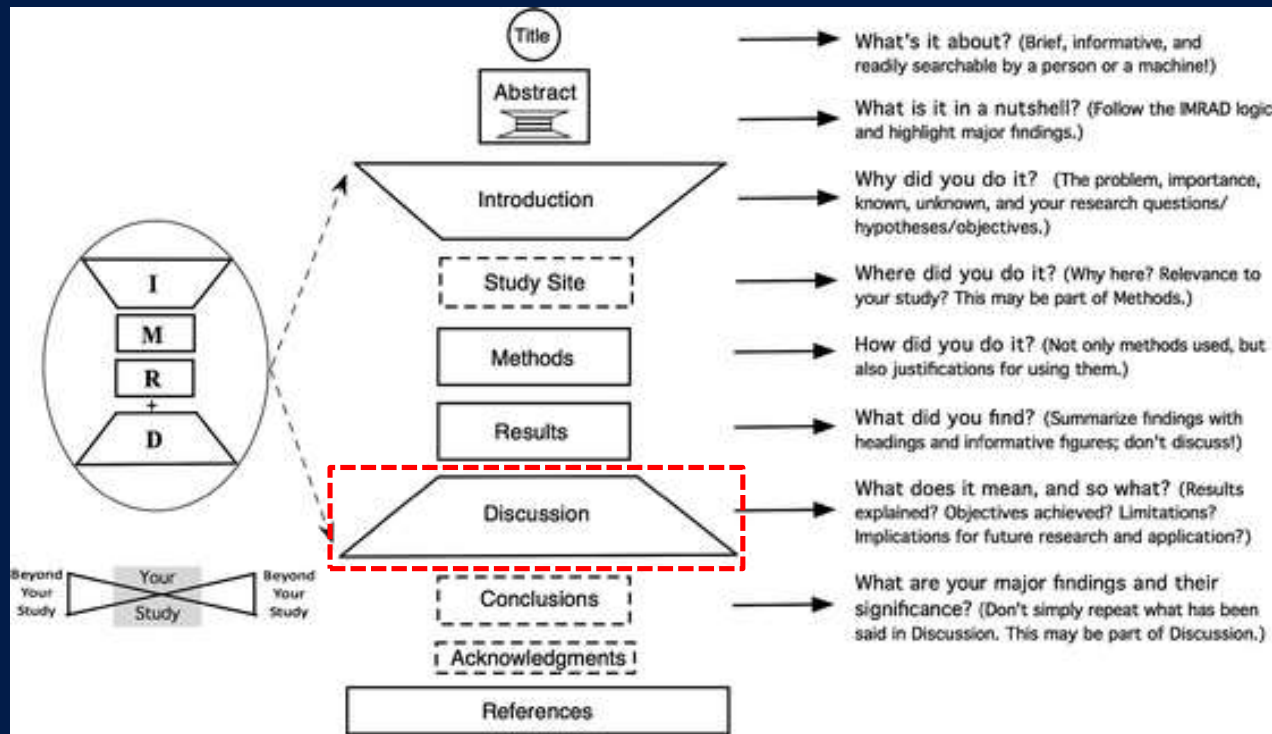
...

Francis Oloo (Dr. rer. Nat)

Outline

- Understand the function of Discussion relative to Results
- Structuring the discussion section
- Link the discussion back to the study's objectives

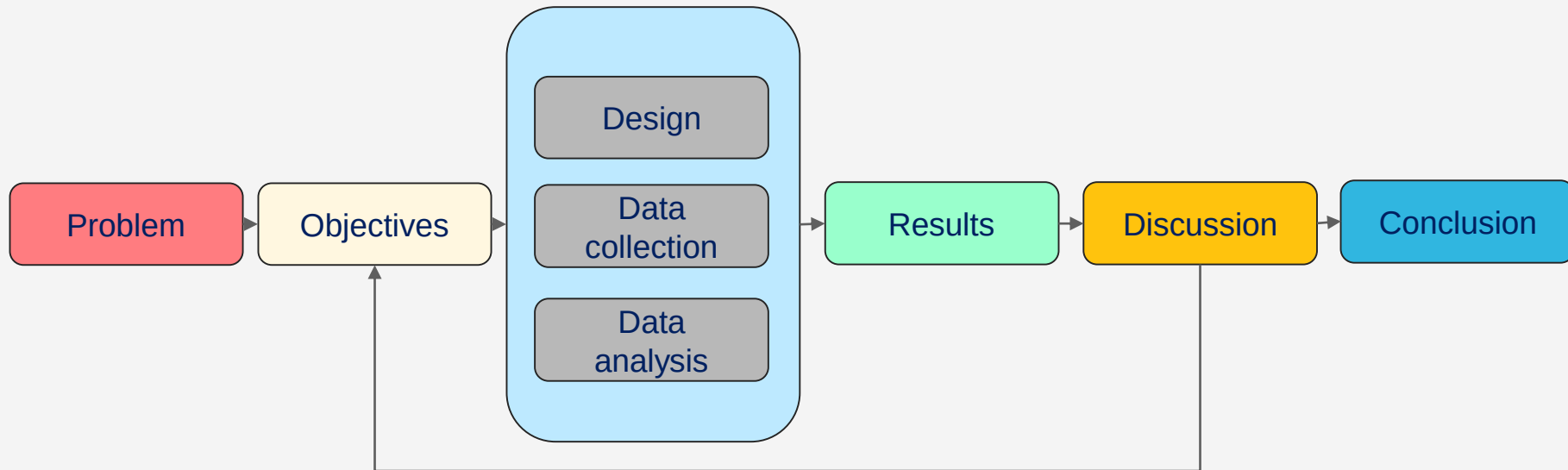
Recap on IMRAD method



Discussion

The introduction narrows down to a gap/objective while Discussion widens back out to the field

Research Workflow



A Discussion that cannot be traced back, point by point, to the study's stated objectives has broken the "golden thread" — even if every sentence in it is well written.

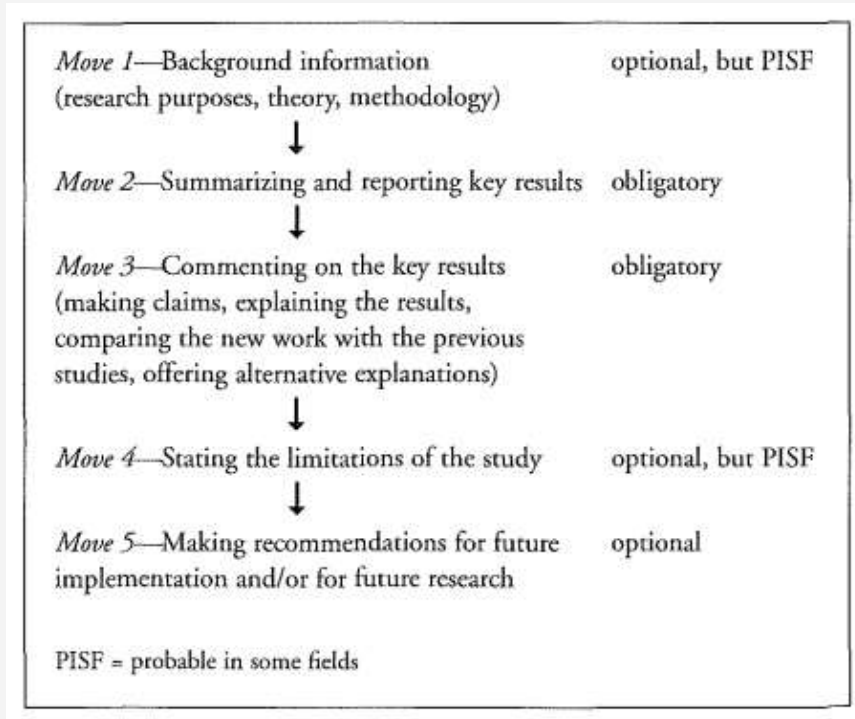
Results → Discussion (**what changes?**)

Results	Discussion
Reports what was found	Interprets what the finding means
Descriptive, neutral tone	Evaluative, argumentative tone
No reference to prior literature	Compares findings to prior literature
No claims beyond the data	Draws implications, with caveats
Organised by variable/theme/hypothesis	Organised by argument, revisiting objectives

Argument

**Arguments are claims backed by reasons that
are supported by evidence**

Writing the discussion



- One typical model is **5-step Move model** from Swales & Feak's (2012) → Get a copy [here](#), page 368

Sample 5-Step Move Model

Move 1 – Background: *This study set out to quantify how a new drought-tolerant maize variety performs relative to the local check variety under farmer-managed conditions, given limited independent field-scale evidence.*

Move 2 – Summarizing key results: *Across all three trial sites, the drought-tolerant variety showed significantly lower yield loss under moderate-to-severe drought stress than the local check.*

Move 3 – Commenting on key results: *This effect size is practically meaningful for smallholder food security and is consistent with variety trials reported elsewhere in the region; the smaller effect observed at one site is likely explained by its shallower soil depth, which limited moisture retention for both varieties.*

Move 4 – Summarizing the study / limitations: *These findings are based on a single growing season across three sites, which limits how confidently the results can be generalized across seasons or agro-ecological zones.*

Move 5 – Recommendations: *Multi-season replication is recommended before any variety-replacement programme is scaled to farmers more broadly.*

Discuss more examples from participants

Reporting, Summarizing, **Commenting**

Reporting

"Yield loss under drought stress was 18% lower for the new variety (95% CI: 12–24%)."

Summarizing

"Across all three sites, the new variety consistently outperformed the local check under drought stress."

Commenting

Interpreting, comparing to literature, accounting for anomalies, evaluating strength/importance

Commenting on Results

Interpreting

Explaining what a result means in plain terms

"This suggests that..."

Comparing

Situating the finding against prior studies

"This finding is consistent with..."

Accounting For

A plausible explanation for an unexpected result

"One possible explanation is..."

Evaluating

Judging the strength or importance of a finding

"This effect, while significant, was modest..."

Linking Discussion to Objectives

- Build one row per objective/question/hypothesis, trace the result and the discussion point
- Anything in the Discussion that can't be traced to a row is either **scope creep** or **unlabeled exploratory material**

Objective / Question / Hypothesis	What Results Showed	Discussion Point to Make
<i>(as stated in Introduction)</i>	<i>(the specific finding)</i>	<i>(interpretation, comparison, implication)</i>

Example from quantitative data

Discussion Element	Content
Background	Restate the objective: quantifying yield difference under drought stress
Reporting/ Summarizing	New variety showed significantly lower yield loss across all three sites
Commenting	Interpret effect as practically meaningful; compares with prior trials; accounts for the weaker site due to shallower soil
Evaluating the study	One growing season in three sites — limits generalisability
Deductions	Recommends multi-season replication before scaling — answers the objective

Handling unexpected results

- State the unexpected result plainly before attempting to explain it
- Offer a plausible account
- Distinguish a null result (no detectable effect) from a disproved hypothesis
- Do not reinterpret the objective so that null result looks plausible

Implications/Recommendations

- **Practical implications** — what practitioners/policymakers might do differently
- **Research implications** — what the field should study next
- **Recommendations** should be proportionate to the evidence (replicate $\neq$ scale nationally)

Common errors

- Re-presenting full results tables instead of interpreting them
- **Introducing new data** or citations without framing them as new context
- Discussing findings out of order from how objectives were introduced
- Claiming an objective was met when results only partially support it
- Generic limitations ("more research is needed") that specify nothing
- Never explicitly stating whether each objective was answered

Assignment

- Using your own study or a sample published work, trace the results and discussion points from the objectives. Write a paragraph of one discussion point using the 5-Step Move model

Objective / Question / Hypothesis	What Results Showed	Discussion Point to Make
<i>(as stated in Introduction)</i>	<i>(the specific finding)</i>	<i>(interpretation, comparison, implication)</i>

Questions and discussions